

N

Model Arithmetic.

↳ Generation time.

→ SOTA generation technique (CTH).
Controlled text

LM $\xrightarrow[\text{text}]{\text{generating}}$ (CTH fashion).

↳ Control over the generation of the output tokens.

Discrete prob. distribution: P
 $P(x)$

DISCOVER
associates a probability $P(x)$
with every x in a finite set
 T .

LM $\rightarrow$ x ? then
 T finite set of tokens.

LM is find the prob. of a
token x_k given all previous

tokens $x_1 \dots x_{k-1}$

$$P(x_k | x_1 \dots x_{k-1}).$$

K-L divergence:

(Kullback-Leibler).

between two prob. distributions P, Q :

$$D_{KL}(P||Q|x_{1:k-1}) = \sum_{x \in T} P(x|x_{1:k-1}) \log \frac{P(x|x_{1:k-1})}{Q(x|x_{1:k-1})}$$

$x_{1:k-1}$ → conditioning over the
last $k-1$ tokens.

$$\rightarrow D_{KL}(P||Q)$$

M
↓

LM: $P(x_k | x_{1:k-1})$ $M(x_k | x_{1:k-1})$

CTG:

Attribute 'a' (style/topic)

Bias the output distribution of M
with respect to 'a'

parameter: λ controls how much we want
to bias the output distribution.

Speculative Sampling X

Model Arithmetic.

Obtain an output distribution: P
instant Distribution

from a set of inputs
 $Q_1, \dots, Q_n$ by minimizing
the linear combination of KLD
 $D_{KL}(P \parallel Q_i)$

Weighted K-L optimality:

$D_{KL}(P \parallel Q) \rightarrow$ attends to each token
"equally"

Certain tokens are generally more
associated with the bias attribute 'a'

└ these tokens
should be preferred
more!

→ Consequently other tokens should be preferred less.

Weighted version of KLD: $D_{KL}^{[f]}$

$$D_{KL}^{[f]}(P \parallel Q | x_{1:k-1}) = \sum_x P(x | x_{1:k-1}) f(x, x_{1:k-1}) \log \frac{P(x | x_{1:k-1})}{Q(x | x_{1:k-1})}$$

$f: \mathcal{T} \times \mathcal{T}^{k-1} \rightarrow \mathbb{R}$ assigns weight to each token.

Weighted KL optimality theorem:

$$\sum_{i=1}^n D_{KL}^{[f_i]}(P \parallel Q_i | x_{1:k-1})$$

$$\arg \min_P \sum_{i=1}^n KL(Q_i \| P)$$

$Q_1, \dots, Q_n$
input distributions.

$$P(x_k = x | x_{1:k-1}) = \frac{1}{\sum_{i=1}^n f_i(x, x_{1:k-1})} \sum_{i=1}^n f_i(x, x_{1:k-1}) \log Q_i(x | x_{1:k-1})$$

Proof of Theorem 1.

- ① $P(x) = P(x_k = x | x_{1:k-1})$, $Q_i(x) = Q_i(x | x_{1:k-1})$
- ② New objective $J(\theta) = \sum_{i=1}^n \sum_x P(x) \frac{f_i(x)}{Q_i(x)} \log \frac{P(x)}{Q_i(x)}$
- ↓
biasing weight.

$$(3) \log \frac{P(x)}{Q_i(x)} = \log P(x) - \log Q_i(x)$$

$$(4) JP = \sum_{i=1}^n \sum_x P(x) f_i(x) (\log P(x) - \log Q_i(x))$$

$$(5) \left(\sum_i \right) \left(\sum_x P(x) f_i(x) \log P(x) \right) - \sum_i \sum_x P(x) f_i(x) Q_i(x)$$

$$(6) F(x) = \sum_i f_i(x)$$

$$(7) JP = \sum_x P(x) F(x) \log P(x) - \sum_x P(x) \sum_i f_i(x) \log Q_i(x)$$

prob. distribution. $\sum_x P(x) = 1$

⑧ P is a probability distribution over x

⑨ minimization of JP has to be done under the constraint $\sum_x P(x) = 1$

⑩ Blend the constraint into the minimization objective.

Lagrangian multiplier:

$$L(P) = JP + \lambda \left(\sum_x P(x) - 1 \right).$$

$$\textcircled{11} L(P) = \sum_x P(x) f(x) \ln P(x) - \sum_x P(x) \sum_i f_i(x) \ln g_i(x) + \lambda \left(\sum_x P(x) - 1 \right)$$

⑪ Partial derivative with respect to

(12)

$$P(x) \Big|_{x=x}$$

Term 1:

$$\frac{\partial}{\partial P(x)} [P(x) F(x) \ln P(x)] = F(x) \ln [1 + P(x)]$$

Term 2:

$$\frac{\partial}{\partial P(x)} \left[-P(x) \sum_i f_i(x) g_i(x) \right] = -\sum_i f_i(x) g_i(x)$$

constraint
term

$$\frac{\partial}{\partial P(x)} \lambda(P(x)) = 0$$

Minimization means:

$$[P(x) + 1] - \sum_i f_i(x) g_i(x) + \lambda = 0$$

$$F(x) \ln F(x) = \sum_i f_i(x) \ln Q_i(x) - F(x) - 1$$

$$\ln P(x) = \frac{\sum_i f_i(x) \ln Q_i(x)}{F(x)} - 1 - \frac{1}{F(x)}$$

$$P(x) = \exp \left(\frac{\sum_i f_i(x) \ln Q_i(x)}{F(x)} \right) \cdot C$$

↓
absorbed
in a
constant
C.

$P(x) \rightarrow$ prob distribution.

$$\int P(x) = 1$$

$$\sum_x P(x)$$

$$P(x) =$$

$$\frac{\exp\left(\frac{\sum_i f_i(x) \log Q_i(x)}{F(x)}\right)}{\sum_y \exp\left(\frac{\sum_i f_i(y) \log Q_i(y)}{F(y)}\right)}$$

$$P(x) = \sigma\left(\frac{1}{\sum f_i(x)} \sum_i f_i(x) \log Q_i(x)\right)$$

$\searrow$ Solⁿ

a) stronger penalty
for toxicity

Model level strength λ_i

AND

More bias
for analysis

Why

Token strength $f_i(x)$
wash like $\swarrow$
kill $\searrow$ hate $\rightarrow$ denied

Softmax is scale invariant. Drop the
normalizing denominator

$$P(x) \propto \sigma \left(\sum_i \lambda_i f_i(x) \theta_i(x) \right)$$

Model arithmetic = weighted sum of logits.
Each word contributes $\theta_i f_i'(x)$ by $\theta_i(x)$
a token score.

✓ to the final form:

E.g. Linear combination:

$$M = \lambda_1 Q_1 + \lambda_2 Q_2$$

Negative bias

$$M = Q_1 - A Q_2$$

Union operator:

Indicator functions:

$$I_1(x) = [Q_1(x) > Q_2(x)]$$

$$I_2(x) = 1 - I_1(x)$$

Optimize

$$D_{KL}^{[I_1]}(P \parallel Q_1) + D_{KL}^{[I_2]}(P \parallel Q_2)$$

sol: $\sigma(\max(\log Q_1, \log Q_2))$

important arithmetic for safety:

$$Q_1 - X \text{ union}(Q_1, Q_2)$$

$$Q_1 > Q_2 \rightarrow \text{behaves like } Q_1$$

$$Q_2 > Q_1$$

Penalty to bias the distribution
away from Q_2